Highlights

Hybrid CNN–LSTM Architecture for EEG-based tDCS Remission Prediction

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- Feature-embedded Hybrid CNN–LSTM with attention achieves 89.3% ROC-AUC and perfect precision (100%) for patient-level tDCS remission prediction
- Controlled ablation reveals baseline correction shifts performance by 15–17% ROC-AUC, comparable to architectural improvements
- Simple tabular MLP matches hybrid model ROC-AUC after baseline correction while achieving perfect recall, supporting complementary screening/confirmation workflow
- Preprocessing sensitivity is first-order: systematic evaluation should precede architectural complexity increases

Hybrid CNN–LSTM Architecture for EEG-based tDCS Remission Prediction

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ABSTRACT

We present a feature-embedded Hybrid 1D Convolutional Neural Network–Long Short-Term Memory (CNN–LSTM) architecture with multi-head attention for predicting transcranial direct current stimulation (tDCS) remission in major depressive disorder (MDD) from four-channel resting-state EEG. The architecture combines convolutional feature extraction, bidirectional LSTM processing, and attention mechanisms to learn discriminative patterns from compact feature vectors. Using leave-one-patient-out cross-validation on 21 MDD patients, the hybrid model achieves 89.3% ROC-AUC with perfect precision (100%) and 71.4% recall, demonstrating high-precision patient-level prediction suitable for clinical confirmation workflows. However, a controlled ablation study reveals a surprising finding: standard baseline correction (per-channel, per-window mean-centering) shifts performance by 15–17% ROC-AUC across both the hybrid architecture and a simple tabular multilayer perceptron (MLP) control. After baseline correction, the simple MLP matches the hybrid model's ROC-AUC (0.898 vs. 0.893) while achieving perfect recall (100%), supporting a complementary screening operating point. These results demonstrate that (i) the hybrid architecture provides valuable high-precision prediction when preprocessing is properly controlled, and (ii) preprocessing sensitivity is a first-order consideration that can outweigh architectural complexity, highlighting the critical importance of systematic preprocessing evaluation in EEG-based clinical prediction.

1. Introduction

Major depressive disorder (MDD) affects over 280 million people worldwide, representing one of the leading causes of disability and contributing significantly to the global burden of disease [40]. According to the World Health Organization, depression is projected to become the top contributor to disease burden by 2030, emphasizing the urgent need for effective treatment approaches [30]. MDD is characterised by prolonged low mood, diminished interest, exhaustion, disrupted sleep patterns, and feelings of unease [5]. The disorder exhibits considerable heterogeneity, with a wide range of symptom clusters and no single underlying aetiology. When diagnosing and treating MDD, it is essential to consider all aspects of the condition, understanding that improvements in certain symptoms such as mood may occur before improvements in other symptoms like sleep disturbance [26].

MDD has a substantial impact on quality of life, specifically in terms of health-related aspects, with treatment response significantly affecting patient outcomes [9]. Current pharmacological treatments have limited effectiveness, resulting in high occurrence of residual symptoms and side effects, emphasizing the necessity for innovative therapies [36]. The development of reliable biomarkers for predicting treatment response represents a critical step toward precision psychiatry, potentially transforming clinical decision-making and improving patient outcomes [21, 20].

1.1. Transcranial Direct Current Stimulation for Depression

Transcranial direct current stimulation (tDCS) is emerging as a promising non-invasive neurostimulation technique, offering significant therapeutic benefits with minimal side effects compared to traditional pharmacological interventions [8]. tDCS delivers low-amplitude direct current (typically 1-2 mA) to the scalp via electrodes, modulating cortical excitability in targeted brain regions. These currents induce neuromodulatory changes in cortical activity that persist beyond the duration of stimulation. In depression treatment, the anode is commonly placed over the left dorsolateral prefrontal cortex with cathodal stimulation on the right hemisphere to rebalance neural activity associated with mood regulation [36].

Recent meta-analyses demonstrate tDCS's clinical efficacy in depression treatment [37]. A meta-analysis of 23 randomized controlled trials involving 1,092 participants found that active tDCS was superior to sham stimulation, with a moderate effect size (Hedges' g $\approx$ 0.46) in depression severity and approximately doubled rates of clinical response (33% vs 16%) and remission (19% vs 10%), while discontinuation rates showed no significant difference between active and sham treatment arms [44].

Abbreviations: MDD, Major Depressive Disorder; tDCS, Transcranial Direct Current Stimulation; EEG, Electroencephalogram; DC, Direct Current; MLP, Multilayer Perceptron; CNN, Convolutional Neural Network; LSTM, Long Short-Term Memory; PSD, Power Spectral Density; AUC, Area Under the Curve; LOPO-CV, Leave-One-Patient-Out Cross-Validation.

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Individual participant interviews have also shown high acceptability for tDCS treatment [46, 45].

A fully remote, multisite, double-blind, placebo-controlled randomized trial of home-based tDCS for MDD (n=174) demonstrated that active stimulation was associated with significantly greater improvements in depressive symptoms compared to sham, with no differences in discontinuation rates [51]. The clinical trial observed no serious adverse events related to tDCS device usage, and no participants developing mania or hypomania. These findings support home-based, remotely supervised tDCS as an efficacious, acceptable, and safe antidepressant intervention [51].

1.2. EEG-Based Biomarkers for Treatment Prediction

Electroencephalography (EEG) offers a non-invasive, cost-effective approach for identifying predictive biomarkers of treatment response. EEG signals serve as reliable indicators of brain health conditions, offering vital insights for the diagnosis of neurological disorders [48]. The capacity to record alterations in cerebral function, along with its straightforwardness and lack of invasiveness, renders EEG a desirable instrument in diverse medical and scientific contexts [32]. Technological improvements, such as the development of wearable EEG devices, have improved the practicality and usefulness of this technology [11].

EEG-based biomarkers offer a means to stratify patients who will likely show clinical response from those with lower likelihood before treatment begins [3]. Investigation of EEG characteristics such as power spectral density, coherence, and fractal dimensions that have predictive value for tDCS treatment response would aid in personalizing treatment [21, 20]. Recent advances in deep learning have enabled sophisticated analysis of EEG signals, with hybrid Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) architectures achieving promising results in various classification tasks [1, 47]. However, most research focuses on developing increasingly complex model architectures, often overlooking the fundamental importance of signal preprocessing.

1.3. Baseline Correction in EEG and Why It Matters for Prediction

EEG recordings contain baseline shifts arising from electrode–skin interfaces, referencing choices, and slow drifts; these effects vary across channels and can change within-session [48]. Baseline correction (including mean-centering or baseline subtraction) is therefore a standard preprocessing step in many EEG pipelines, yet its downstream impact on feature computation and classification can be substantial. Spectral analysis and time–frequency transforms are sensitive to baseline shifts [35], and window-wise baseline variation can bias feature distributions and impair cross-subject generalization. While high-pass filtering partially mitigates slow drift, per-channel offsets can persist within analysis windows—particularly in portable/home EEG—and can interact with normalization and feature selection. In this work we treat per-channel, per-window mean subtraction as a representative baseline-correction choice and evaluate its impact explicitly via controlled ablation.

The importance of artifact correction and precise preprocessing techniques in improving EEG data quality has been emphasized in the literature [29]. However, systematic studies comparing the impact of DC offset removal against architectural improvements in EEG classification are notably absent. Most preprocessing pipelines either apply global normalization or rely on filtering alone, potentially leaving channel-specific and window-specific offsets that could degrade model performance. This gap is particularly significant given recent evidence that simpler models with careful feature engineering can achieve strong results [53], suggesting that sophisticated architectures may not always be necessary when data quality is ensured.

1.4. Research Questions

This study addresses four key research questions:

1. **RQ1**: How sensitive is patient-level tDCS remission prediction to standard baseline correction (per-channel, per-window centering)?
2. **RQ2**: Does the feature-embedded Hybrid CNN–LSTM provide added value beyond preprocessing when evaluated under LOPO patient-wise testing?
3. **RQ3**: Do simple and complex models exhibit different clinically relevant operating points (screening vs. confirmation) under the same preprocessing?
4. **RQ4**: What practical recommendations for deployable EEG prediction pipelines follow when preprocessing and architecture are evaluated jointly?

1.5. Contributions

This work makes the following contributions:

- **Architecture-led contribution**: We present a compact feature-embedded Hybrid CNN–LSTM with lightweight attention tailored to low-density EEG feature vectors, enabling high-precision patient-level prediction under LOPO evaluation.
- **Controlled sensitivity analysis**: We quantify how a standard baseline-correction choice (per-window mean-centering) shifts performance across a simple and a hybrid model, promoting reproducible reporting of preprocessing sensitivity rather than attributing gains solely to architecture.
- **Clinical operating-point insight**: We identify complementary roles for a high-recall screening model (tabular MLP) and a high-precision confirmation model (hybrid), suggesting a pragmatic decision-support workflow.

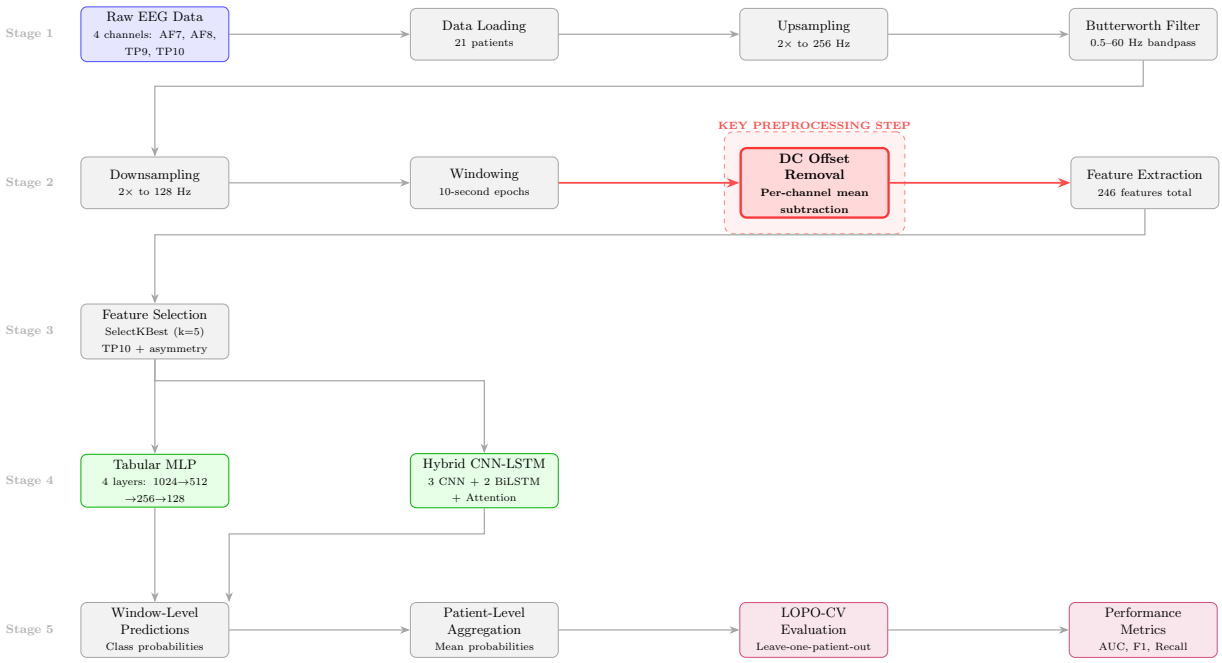

Figure 1: Overview of the EEG processing and classification pipeline. Raw EEG from 4 channels (AF7, AF8, TP9, TP10) undergoes filtering, windowing, and a standard baseline-correction step (per-channel, per-window mean-centering) that we evaluate via controlled ablation. Extracted features (SelectKBest, k=5) are then fed to either a simple tabular MLP (control) or a feature-embedded Hybrid CNN–LSTM with attention, with window-level predictions aggregated to patient-level classifications under LOPO cross-validation.

- **Deployment relevance**: We demonstrate clinically meaningful prediction using a four-electrode montage and a small feature set, supporting translation to portable/home settings [49].

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on EEG preprocessing, deep learning for EEG classification, and depression treatment prediction. Section 3 outlines the methodology, including the dataset, DC offset removal algorithm, model architectures, and evaluation protocol. Section 4 presents experimental results comparing preprocessing effects across model architectures. Section 5 discusses the implications of our findings for clinical EEG applications. Finally, Section 6 concludes with recommendations for practitioners.

Figure 1 provides an overview of our experimental pipeline, including the baseline-correction step that we evaluate explicitly via ablation.

2. Related Work

2.1. EEG Preprocessing for Clinical Applications

Standard EEG preprocessing pipelines typically include filtering, artifact removal, and normalization [29]. EEG data is analysed using a range of mathematical tools and techniques, such as spectral analysis and wavelet transform, which aid in identifying distinct patterns [35]. Bandpass filtering (commonly 0.5-60 Hz) removes slow drifts and high-frequency noise, while artifact removal techniques address eye movements, muscle activity, and other physiological contamination.

Several studies have noted the importance of proper baseline correction in EEG analysis [10], but systematic comparisons of DC offset removal effects on classification performance are rare. Mahato et al. [29] emphasized the importance of artifact correction and precise preprocessing techniques in improving the quality and dependability of EEG data for detecting depression, suggesting that subsequent investigations should prioritize enhancement of artifact removal techniques and examine their influence on classification accuracy and resilience. Comprehensive artifact correction approaches, including eye movement artifact removal using independent component analysis combined with eye tracking, have been developed [42], but channel-specific DC offsets within analysis windows often receive insufficient attention, with most work focusing on more complex artifact removal techniques while assuming basic preprocessing is adequate.

The development of wearable EEG devices has improved the practicality of EEG technology [11], but portable systems with reduced electrode counts may be more susceptible to DC offset issues due to simplified electrode configurations. The capacity to record alterations in cerebral function, along with its straightforwardness and lack of invasiveness, renders EEG a desirable instrument in diverse clinical contexts [32], provided that appropriate preprocessing ensures signal quality.

2.2. Deep Learning for EEG Classification

Deep learning approaches for EEG analysis have evolved from simple convolutional networks to sophisticated hybrid architectures. Classical approaches using support vector machines (SVM), linear discriminant analysis (LDA), and extreme learning machines (ELM) have been used to classify EEG signals for tDCS response prediction [2]. Mumtaz et al. [34] proposed wavelet-based techniques to predict antidepressant outcomes, reporting classification accuracy of 87.5% using logistic regression on wavelet features.

More recent work has employed hybrid architectures combining convolutional and recurrent components. Adamson et al. [1] used a 1D CNN followed by an LSTM for rTMS response classification, reporting very high accuracies (up to 99% in sex-specific models). Shahabi et al. [47] combined wavelet time-frequency representations with convolutional recurrent networks and attention, reporting 80-90% accuracy across private and public cohorts, with LSTM-based sequence modeling improving over static CNN features.

A distinguishing characteristic of EEG signals is their rich temporal structure, reflecting dynamic brain oscillations and events. Traditional EEG biomarkers for depression response have often been computed as static features such as average band power, which may overlook how brain activity evolves over time [23]. Temporal sequence analysis techniques can potentially detect complex temporal patterns that distinguish responders from non-responders. Recent advances in deep learning have made it possible to apply temporal sequence models to EEG data to automatically learn discriminative patterns that human observers or simpler algorithms might miss [39]. Attention-based architectures have achieved state-of-the-art performance on single-channel EEG classification tasks such as sleep staging, demonstrating the potential of these approaches for clinical EEG applications [18].

Additional investigation has explored nonlinear features of EEG and their potential in predicting treatment outcomes. These features can capture more intricate aspects of brain activity linked to treatment response [16]. Deep autoencoders and convolutional networks have been applied to unsupervised feature learning from EEG signals, achieving high classification accuracy through automatic feature extraction [50]. These approaches suggest that learned representations may capture discriminative information beyond hand-crafted features, though the relative contribution of preprocessing versus architecture remains unclear.

2.3. Data Quality versus Model Complexity

The machine learning principle of "garbage in, garbage out" emphasizes data quality importance, yet systematic studies comparing preprocessing effects against model complexity in EEG applications are limited. Recent work comparing CNN-extracted features with hand-crafted features in activity recognition found that while CNNs can automatically learn discriminative representations, their performance depends heavily on architecture parameters and input quality [12]. Existing EEG studies often employ high-density electrode systems with 19-128 channels, creating significant barriers to clinical deployment due to complexity, cost, and patient compliance issues [49]. The development of effective prediction models using reduced electrode configurations represents a critical need for translating research findings into clinical practice.

Wu et al. [53] proposed Sparse EEG Latent Space Regression (SELSER), using a broad spectral feature set to predict antidepressant outcomes as a regression task, reporting cross-validated R^2 up to 0.60 on a large trial dataset (309 patients) with independent-sample testing. Zhdanov et al. [58] trained an SVM on baseline EEG from 122 escitalopram patients using multiple feature families, reporting classification accuracy of ∼79% for responder prediction; incorporating early-treatment EEG increased accuracy to ∼82%. These results suggest that careful feature engineering and data quality may be as important as model complexity.

To enhance reliability and validity of future research, it is crucial to address limitations such as the absence of sham conditions in certain studies, limited electrode arrays, and small sample sizes [56]. Integration of data from different modalities using machine learning methods may achieve more comprehensive prediction of treatment response [34].

2.4. Depression Remission Prediction

EEG biomarkers for depression treatment response include frontal alpha asymmetry, theta cordance, and various spectral and connectivity features [25]. Puthankattil and Joseph [43] and Khadidos et al. [28] emphasize the importance of frequency-domain characteristics, such as Relative Wavelet Energy and band power analysis, in differentiating EEG signals between normal and depressive states. Ghiasi et al. [22] and Zhao et al. [57] emphasize the significance of spatial information and microstate analysis in understanding EEG characteristics in individuals with depression.

Beyond single-modality markers, converging evidence suggests that MDD comprises distinct neuroanatomical dimensions with different treatment propensities, identified through structural MRI using semi-supervised clustering [19]. This dimensional framing provides context for integrating EEG with other modalities to stratify patients and test whether EEG predictors express differently across neuroanatomical dimensions.

Prior work on the same clinical dataset achieved 85.7% accuracy using CNN-LSTM architectures with phase-locking value connectivity features [55]. Across home-tDCS cohorts in unipolar MDD and bipolar depression, low-density PLV connectivity features with deep learning have achieved ∼69-72% accuracy [55, 54]. Moncy et al. [33] applied end-to-end deep models on PSD vectors from the same four-channel EEG dataset, achieving

85.7% accuracy with a sensitivity-oriented operating point. However, the relative contributions of preprocessing and architecture in these results remain unclear, motivating the present systematic comparison.

A meta-analysis of EEG-based treatment response prediction across 15 studies with 479 patients reported area under the curve of 0.91, sensitivity of 83%, and specificity of 86% across various treatment modalities [13]. These benchmarks provide context for evaluating our preprocessing-focused approach.

3. Methodology

3.1. Participants

This study utilized EEG data from participants diagnosed with major depressive disorder who participated in a fully remote, multisite, double-blind, placebo-controlled, randomized superiority trial investigating home-based tDCS for treating MDD [51]. The study was conducted by the University of East London School of Psychology in collaboration with clinical partners. All participants provided written informed consent. Ethical approval was granted by the South Central–Hampshire B Research Ethics Committee (UK). Comprehensive clinical and demographic details for the trial are reported in Woodham et al. [51].

Participants were adults aged 18 years or older, diagnosed with MDD and in a current depressive episode of moderate severity or higher, as indicated by a score of 16 or more on the 17-item Hamilton Depression Rating Scale (HDRS), following DSM-5 criteria. Recruitment encompassed regions in England, Wales, and Texas, USA. Inclusion criteria required participants not to be resistant to depression treatments or on medications affecting cortical excitability.

Exclusion criteria included inadequate response to antidepressant treatments, significant suicide risk, comorbid psychiatric disorders, or use of medications influencing cortical excitability. Written informed consent and access to video conference for remote session supervision were mandatory requirements.

The study demographic comprised 21 participants who completed the full protocol and had usable EEG data. Following the 10-week tDCS intervention period, participants were classified into remission (n=7, 33.3%) and non-remission (n=14, 66.7%) groups based on standardized clinical assessment criteria. This distribution reflects typical clinical response rates observed in tDCS trials for depression treatment [44]. Table 1 summarizes the dataset characteristics.

Table 1
Dataset Characteristics

Characteristic	Value
Participants	21 patients
Non-Remission	14 (66.7%)
Remission	7 (33.3%)
Gender (Female/Male)	18/3
Age (years, mean ± SD)	37.1 ± 9.7
EEG Channels	4 (AF7, AF8, TP9, TP10)
Channel Types	Frontal, Temporal
Sampling Rate	256 Hz
Window Size	10 seconds
Total Windows	1,203
Positive Windows	393 (32.7%)
Negative Windows	810 (67.3%)
Class Ratio	2.06:1 (negative:positive)
Cross-Validation	Leave-One-Patient-Out (21 folds)
Evaluation Level	Patient-level

Table 2
Tabular MLP Architecture

Component	Specification
Input Features	5
Hidden Layers	[1024, 512, 256, 128]
Activation	ReLU
Batch Normalization	After each hidden layer
Dropout Rate	0.3
Output	Binary (2 classes)
Optimizer	AdamW (lr=0.001, wd=0.01)
LR Schedule	Cosine annealing warm restarts
Batch Size	1024
Label Smoothing	0.05

3.2. EEG Data Acquisition and Preprocessing

EEG data were collected during resting state conditions with participants' eyes closed, using a standardized four-electrode configuration (AF7, AF8, TP9, TP10) at a sampling rate of 256 Hz. The EEG signals were recorded in 10-second epochs, with some participants having recordings of 10 minutes duration while others had data split into two 5-minute files.

The preprocessing pipeline used custom Python scripts and standard signal processing libraries. To standardize the data and correct slight window-size inconsistencies, all segments were resampled to a uniform length. Signals were then high-pass filtered at 0.5 Hz to remove slow drifts and low-pass filtered at 60 Hz to suppress high-frequency artifacts using a Butterworth filter. As part of the resampling process to ensure uniform segment lengths, each channel was upsampled, passed through an anti-aliasing low-pass filter, and downsampled to the target length. This resampling step ensures all windows have identical dimensions for batch processing.

EEG window rejection used the device-provided Horse Shoe Indicator (HSI), which scores electrode contact quality per sample (1=good, 2=average, 4=poor). For each 10-second window, we averaged HSI values over samples (per electrode) and retained windows only when the average HSI was <2 (i.e., better than "average" contact); all other windows were discarded to avoid poor-contact artifacts.

Table 3
Hybrid CNN-LSTM Architecture

Component	Specification
Feature Embedding	Linear(5 → 128) + BN + ReLU
Projection & Reshape	Linear(128 → 64), reshape to pseudo-sequence (len=64)
CNN Blocks	3 sequential blocks with temporal reduction
Block 1	[64, 64] filters, k=[5, 3], pool=2 (64→32)
Block 2	[128, 128] filters, k=[3, 3], pool=2 (32→16)
Block 3	[256, 128] filters, k=[3, 1], pool=1 (16→16)
CNN Dropout	[0.1, 0.2, 0.3] progressive spatial dropout
Feature Pyramid	Multi-scale taps: GAP_1 (64), GAP_2 (128), GAP_3 (128)
LSTM Layers	2 bidirectional (operate on seq. len=1)
Layer 1	128 units/direction, return sequences, dropout=0.2/0.1
Layer 2	64 units/direction, final state, dropout=0.3/0.2
Attention	Multi-head (8 heads, dim=64)
Positional Encoding	Enabled
Feature Fusion	Concat[LSTM-attention, feature pyramid, handcrafted skip]
Handcrafted Skip	5 selected features bypass CNN/LSTM
Dense Layers	[256, 128] units
Dense Activation	GELU
Dense Dropout	[0.3, 0.4]

3.3. Baseline Correction (Per-window Mean-centering) Implementation

We implement per-channel, per-window baseline correction via mean subtraction (DC centering), a standard operation in EEG preprocessing. Algorithm 1 describes the procedure and the exact handling of edge cases used in our experiments.

Algorithm 1 Per-Channel, Per-Window DC Offset Removal

Require: EEG window W with C channels
Ensure: DC-corrected window W'

```
for each channel c = 1 to C do
    signal ← W[c]
    nonzero_mask ← signal ≠ 0
    if any zeros in signal then
        fill_value ← mean(signal[nonzero_mask])
        signal[!nonzero_mask] ← fill_value
    end if
    dc_offset ← mean(signal)
    W'[c] ← signal − dc_offset
end for
return W'
```

Key Design Decisions:

- **Per-Channel Processing**: Each channel processed independently to account for electrode-specific DC offsets.
- **Per-Window Processing**: DC offset computed per window to handle temporal drift.
- **Zero Handling**: Exact zeros replaced with mean of non-zero values before centering to prevent bias.
- **Mean Subtraction**: Used over median for consistency with common baseline-correction practice.

Figure 2 illustrates the effect of DC offset removal on EEG signals from all four channels in a representative example.

3.4. Model Architectures

We evaluated two architectures with deliberately different complexity levels.

3.4.1. Tabular MLP (Control Model)

The Tabular MLP serves as a controlled experiment to isolate DC offset removal effects. Table 2 describes the architecture.

3.4.2. Hybrid 1D CNN-LSTM (Complex Model)

The Hybrid model combines convolutional and recurrent components with attention. The architecture processes tabular EEG features through an embedding layer, reshapes them into a pseudo-temporal sequence for convolutional processing, applies multi-scale feature extraction, and combines learned representations with handcrafted features via a skip connection. Table 3 describes the architecture.

3.4.3. Architecture Changes Between Experiments

The Hybrid model experiments included additional changes beyond DC offset removal. The *Baseline* architecture (matching prior work on this dataset) used: ReLU activation in dense layers, 4 attention heads with key dimension 32, and no positional encoding. The *Enhanced* condition includes:

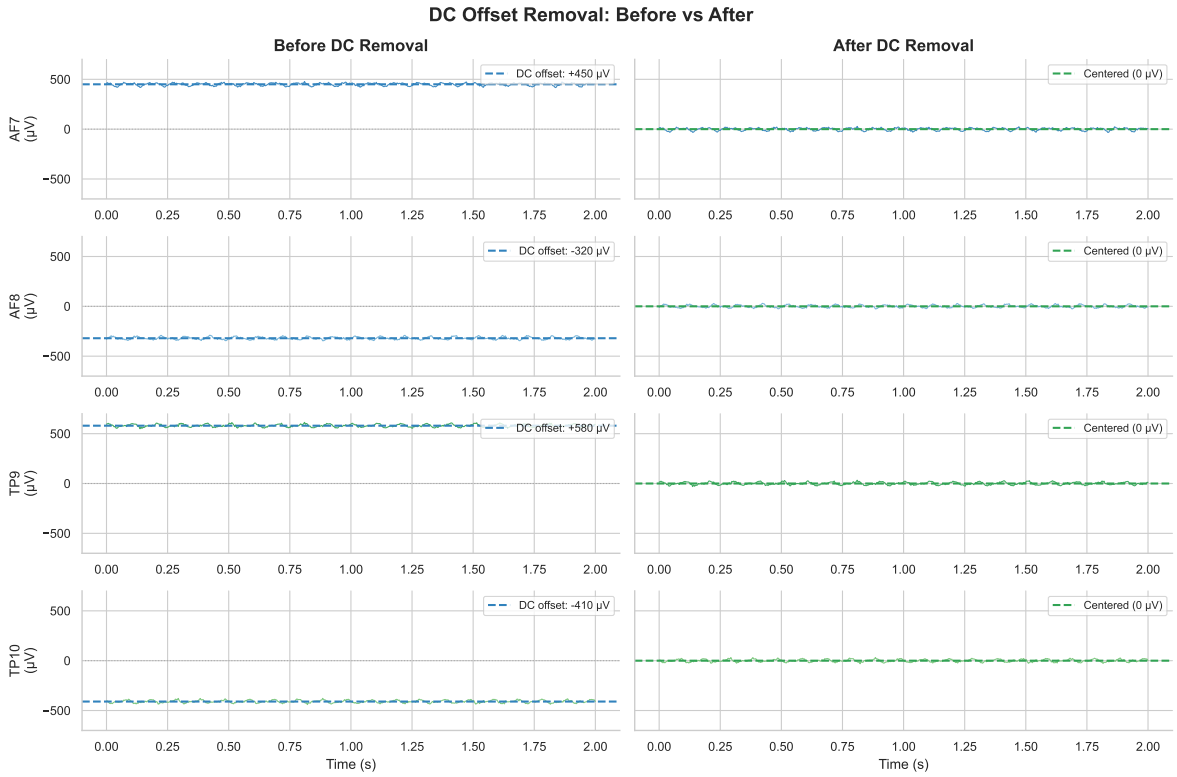

Figure 2: Visualization of DC offset removal effect across all four EEG channels (AF7, AF8, TP9, TP10). Top panels show raw signals with DC offsets creating artificial baselines; bottom panels show the same signals after per-channel mean subtraction, centered around zero. The DC offset magnitudes vary across channels due to electrode-skin interface differences.

- Activation: ReLU → GELU (in dense layers)
- Attention: 4 → 8 heads, 32 → 64 dim
- Positional Encoding: Disabled → Enabled
- Deterministic Training: Enabled for reproducibility

These confounded changes limit attribution of improvements to specific factors in the Hybrid experiments, which is why the Tabular MLP serves as the primary controlled comparison. Note that the LSTM and attention layers operate on a sequence of length 1 (after global average pooling over the pseudo-temporal dimension), functioning primarily as gated channel-wise mixers rather than long-range temporal models.

Figure 3 provides a visual comparison of the two model architectures, illustrating the substantial complexity difference between the simple Tabular MLP and the Hybrid CNN-LSTM.

3.5. Feature Extraction

We extracted features per channel across several signal domains using a configurable pipeline that emphasizes reproducible, lightweight measures aligned with our four-electrode montage. Features span spectral, temporal, entropy/complexity, asymmetry, and cross-hemispheric domains.

Spectral Features: Spectral features were derived from per-channel power spectral densities (PSDs) computed via Welch's method. For each canonical band (delta: 0.5-4 Hz, theta: 4-8 Hz, alpha: 8-13 Hz, beta: 13-30 Hz, gamma: 30-60 Hz), we computed absolute band power and total power, then derived relative band powers by normalizing each band by total power. We further computed frequency-weighted power, spectral entropy from the PSD, spectral edge frequencies (SEF-90 and SEF-95), and PSD summary statistics (mean, standard deviation, maximum, minimum, median, skewness, and kurtosis).

Temporal Features: Time-domain features included basic statistics per channel (mean, standard deviation, variance, skewness, kurtosis, root-mean-square, peak-to-peak amplitude, zero-crossings, and mean absolute deviation) alongside Hjorth parameters (activity, mobility, and complexity) computed from the signal and its derivatives.

Nonlinear and Entropy Features: Nonlinear and entropy-oriented measures were computed per channel to capture complexity beyond second-order statistics. We calculated sample entropy and spectral entropy, as well as correlation dimension, Hurst exponent, and detrended fluctuation analysis (DFA).

Asymmetry Features: Given the importance of hemispheric balance in depression [24], we computed per-band log-power asymmetry for frontal (AF7–AF8) and

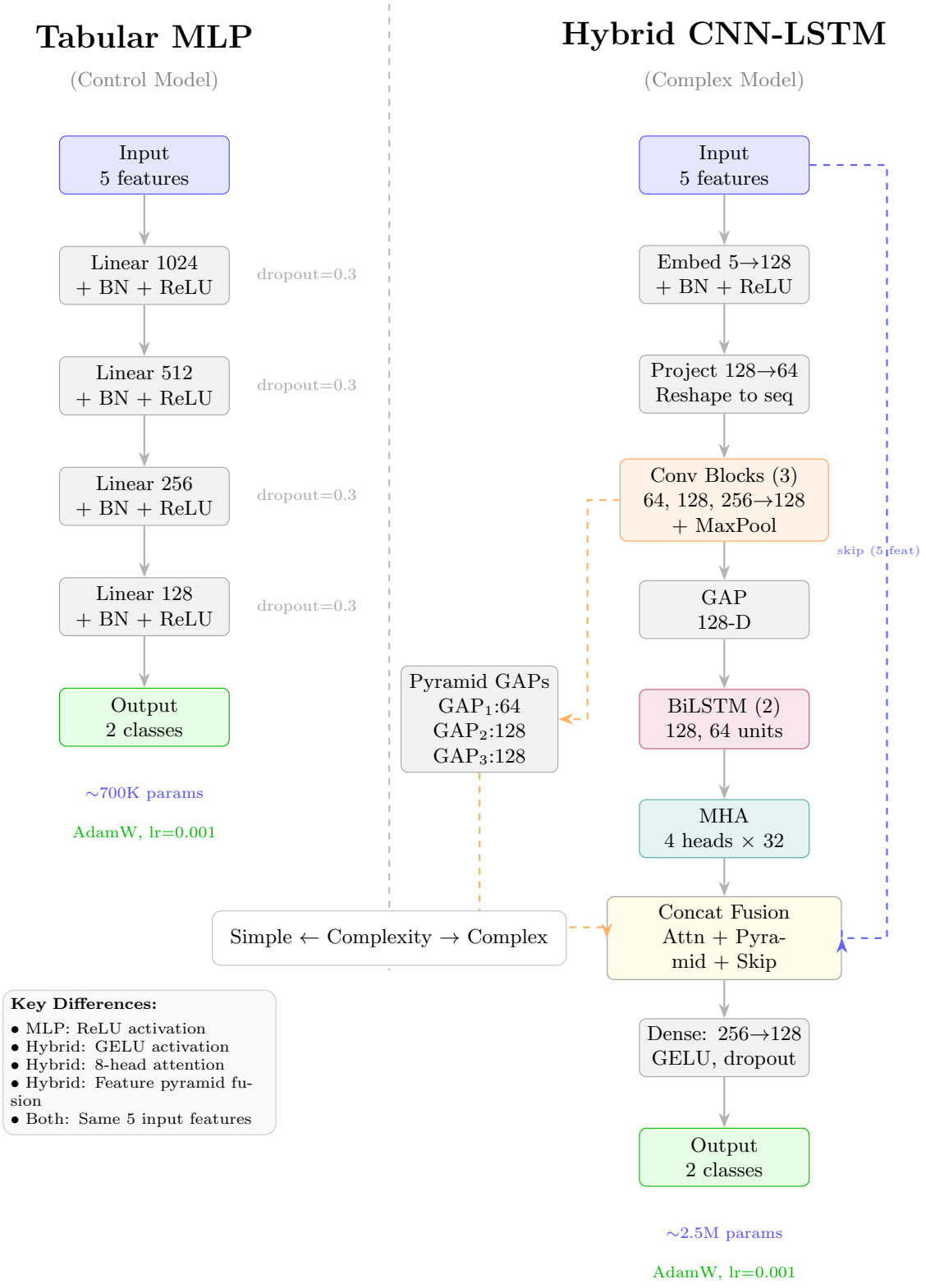

Figure 3: Comparison of model architectures. **Left**: The Tabular MLP is a simple feed-forward network with 4 hidden layers (1024→512→256→128), batch normalization, and dropout. **Right**: The Hybrid CNN-LSTM embeds tabular features (5→128→64), reshapes to a pseudo-temporal sequence, processes through 3 CNN blocks with multi-scale feature pyramid taps, applies bidirectional LSTM and multi-head attention (operating on sequence length 1), and fuses learned features with handcrafted features via skip connection before dense layers. The Tabular MLP has ~700K parameters while the Hybrid model has ~2.5M parameters, yet both achieve similar ROC-AUC (~0.89) after DC offset removal.

temporal (TP9–TP10) pairs, along with ratio-based asymmetries. Cross-hemispheric features included frontal-temporal ratios and differences on each side.

Inter-channel Correlation Features: Between-channel summaries were computed for predefined channel pairs using Pearson correlation coefficient and maximum absolute cross-correlation as proxies for coupling.

3.6. Feature Selection

In total, 246 features were extracted per window. Feature selection used SelectKBest with the ANOVA F-statistic (f_classif) as the scoring function. We evaluated different values of k (k=5 primary, k=20 exploratory) and also a no-selection condition (all 246 features). The k=5 configuration consistently yielded the best downstream patient-level performance and most clinically desirable error profile across models.

The top 5 selected features, ranked by F-statistic (descending order), were:

1. `left_frontal_temporal_diff_beta` (F=150.43)
2. `tp10_psd_std` (F=137.28)
3. `tp10_psd_max` (F=136.26)
4. `tp10_total_power` (F=128.94)
5. `tp10_psd_mean` (F=128.61)

Notably, 4 of 5 features originated from the TP10 (right temporal) electrode, suggesting this region's particular importance for tDCS remission prediction. The fifth feature captures beta-band asymmetry between left frontal and temporal regions, consistent with prior work on hemispheric imbalance in depression [24]. This dominance of TP10 features aligns with prior findings on the same dataset that identified right temporal activity as informative for tDCS response [33]. Figure 4 shows the F-statistics for the top selected features.

3.7. Training and Evaluation

3.7.1. Cross-Validation Strategy

Leave-One-Patient-Out (LOPO) cross-validation was used with 21 folds. In each fold, one patient was held out exclusively for testing while the model trained on remaining patients. An internal validation set from training patients guided early stopping.

3.7.2. Window-to-Patient Aggregation

Patient-level predictions were computed by averaging window-level probabilities:

$$P_{patient} = \frac{1}{N} \sum_{i=1}^{N} P_{window,i} \tag{1}$$

where N is the number of windows for that patient. Binary classification used a threshold of 0.5.

3.7.3. Evaluation Metrics

Primary metrics (patient-level): ROC-AUC, F1-Score, Recall, Precision, Accuracy. Confusion matrices provide detailed error analysis with true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

4. Results

4.1. Experiment 1: Baseline correction ablation (Tabular MLP control)

Table 4 presents the Tabular MLP results without versus with baseline correction (per-channel, per-window mean-centering), with identical architecture in both conditions.

Key Findings: The impact of baseline correction was substantial and consistent across all metrics. Most notably:

- **Perfect Recall**: All 7 positive patients correctly identified (100% sensitivity), eliminating false negatives entirely
- **False Negatives Eliminated**: Reduced from 2 to 0, critical for clinical screening applications
- **ROC-AUC Improvement**: 15.7% relative improvement, from moderate (0.776) to excellent (0.898) discrimination
- **F1 Improvement**: 23.5% relative improvement, reflecting balanced performance gains
- **No Precision Trade-off**: Precision improved slightly despite perfect recall, contradicting the common assumption that recall improvements come at precision cost

These results exceed the benchmarks reported in a recent meta-analysis of EEG-based treatment response prediction, which found average sensitivity of 83% and specificity of 86% across 15 studies [13]. Our Tabular MLP with baseline correction achieved 100% sensitivity with 78.6% specificity (11/14 true negatives), demonstrating that explicit preprocessing control can yield clinically valuable screening behavior.

Table 5 shows the confusion matrices, illustrating the elimination of false negatives.

4.2. Experiment 2: Hybrid model performance under preprocessing and architecture changes

Table 6 presents the Hybrid CNN–LSTM results under a baseline condition and an enhanced condition. The *Baseline* condition uses the original preprocessing pipeline (without explicit per-window mean-centering) and the original architecture configuration (ReLU activation, 4 attention heads with key dimension 32, no positional encoding). The *Enhanced* condition includes per-channel, per-window baseline correction plus architectural improvements (GELU activation, 8 attention heads with key dimension 64, positional encoding enabled).

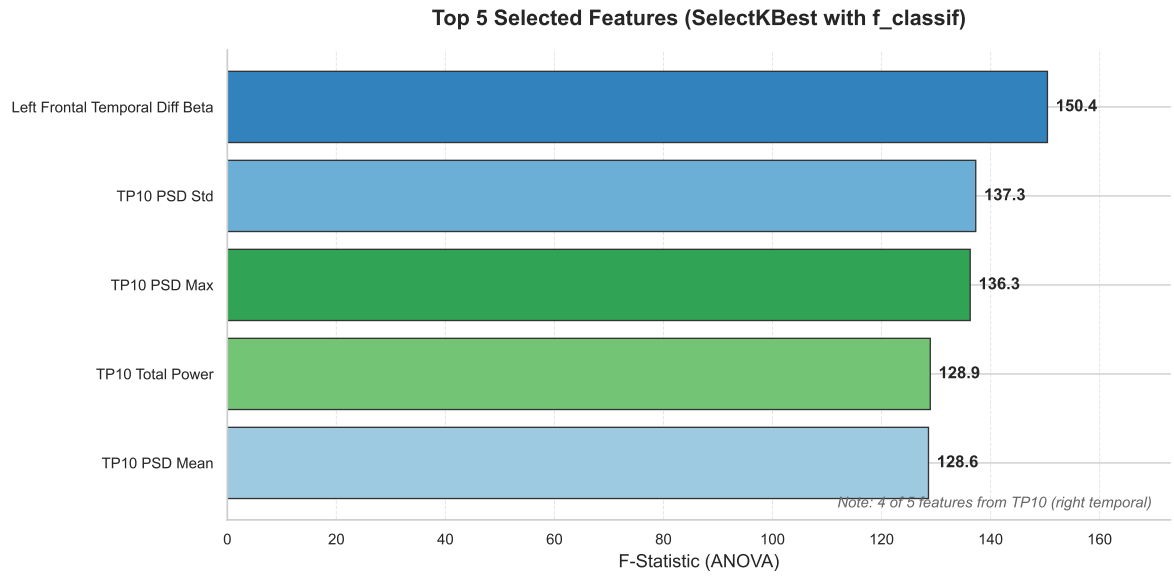

Figure 4: Feature importance ranking based on ANOVA F-statistic (f_classif). The top 5 features selected for classification are dominated by TP10 electrode features (4 of 5), with one asymmetry feature capturing left frontal-temporal beta-band differences. Higher F-statistics indicate greater discriminative power between remission and non-remission classes.

Table 4
Tabular MLP Performance: Effect of baseline correction (per-window mean-centering)

Condition	ROC-AUC	F1-Score	Recall	Precision	Accuracy	FN
Without baseline correction	0.776	0.667	0.714	0.625	0.762	2
With baseline correction	**0.898**	**0.824**	**1.000**	0.700	**0.857**	**0**
Improvement	+0.122	+0.157	+0.286	+0.075	+0.095	-2
% Change	+15.7%	+23.5%	+40.0%	+12.0%	+12.5%	-100%

Table 5
Tabular MLP Confusion Matrices (Patient-Level)

Condition	TN	FP	FN	TP
Without baseline correction	11	3	2	5
With baseline correction	11	3	**0**	**7**

Key Findings: The Hybrid CNN–LSTM showed similar preprocessing sensitivity to the Tabular MLP:

- **Similar ROC-AUC Gain**: Comparable improvement (∼0.13) to Tabular MLP, suggesting baseline correction benefits are broadly architecture-agnostic
- **Maintained Perfect Precision**: No false positives in either condition, reflecting the Hybrid model's more conservative prediction pattern
- **Improved Recall**: Caught 1 additional positive patient (4 → 5 out of 7), though still missing 2 remitters

Interestingly, the Hybrid CNN–LSTM, despite its substantially greater architectural complexity (∼2.5M parameters vs. ∼700K), did not achieve the perfect recall attained by the simpler Tabular MLP. This supports the interpretation that architecture and preprocessing jointly determine the achievable operating point, and that added complexity does not automatically yield higher sensitivity in small clinical datasets.

Table 7 shows the confusion matrices.

Figure 5 provides a visual comparison of the confusion matrices across both models and conditions, highlighting the elimination of false negatives in the Tabular MLP after baseline correction.

4.3. Comparative Analysis

Table 8 compares both models after improvements.

Key Insights: Comparison of the two models after baseline correction reveals several important findings with implications for clinical deployment:

1. **Simpler Model Achieved Higher Recall**: The Tabular MLP caught all positive cases (7/7, 100% recall) while the more complex Hybrid CNN-LSTM missed 2 (5/7, 71.4% recall). This is clinically significant because missing potential treatment responders delays appropriate intervention.
2. **Similar ROC-AUC**: Both models achieved ∼0.89-0.90 after improvements, suggesting that

Table 6
Hybrid CNN–LSTM Performance: baseline vs. enhanced condition (baseline correction plus architecture changes)

Condition	ROC-AUC	F1-Score	Recall	Precision	Accuracy	FN
Baseline (no baseline correction)	0.765	0.727	0.571	1.000	0.857	3
Enhanced (with baseline correction)	**0.893**	**0.833**	0.714	1.000	**0.905**	2
Improvement	+0.128	+0.106	+0.143	0.000	+0.048	-1
% Change	+16.7%	+14.6%	+25.0%	0%	+5.6%	-33.3%

Table 7
Hybrid CNN–LSTM Confusion Matrices (Patient-Level)

Condition	TN	FP	FN	TP
Baseline (no baseline correction)	**14**	**0**	3	4
Enhanced (with baseline correction)	**14**	**0**	2	5

Table 8
Model Comparison After Improvements

Metric	Tabular MLP	Hybrid CNN-LSTM
ROC-AUC	**0.898**	0.893
F1-Score	0.824	**0.833**
Recall	**1.000**	0.714
Precision	0.700	**1.000**
Accuracy	0.857	**0.905**
Complexity	Simple	Complex
Parameters	∼700K	∼2.5M

overall discriminative ability is comparable regardless of architecture when preprocessing is adequate.

3. **Preprocessing sensitivity is first-order**: baseline correction shifts ROC-AUC by ∼0.12–0.13 in both models, comparable to typical architecture tweaks.
4. **Complementary operating points**: the simple model supports screening (high recall), while the hybrid supports confirmation (high precision).

Figure 6 provides a visual comparison of all performance metrics across models and conditions.

Figure 7 summarizes the relative improvements achieved by each model, demonstrating that baseline correction provides substantial gains across all metrics.

4.4. Clinical Interpretation

The two models exhibit different error profiles with distinct clinical implications:

Tabular MLP (With baseline correction):

- True Positives: 7/7 (100% recall)
- True Negatives: 11/14 (78.6% specificity)
- **Clinical Role**: Optimal for screening, catches all remission patients

Hybrid Model (Enhanced):

- True Positives: 5/7 (71.4% recall)
- True Negatives: 14/14 (100% specificity)
- **Clinical Role**: Optimal for confirmation, no false positives

5. Discussion

5.1. Summary of Findings

Our experiments revealed three key findings: (1) baseline correction (per-channel, per-window mean-centering) improved ROC-AUC by 15–17% across both model architectures; (2) a simple Tabular MLP achieved perfect recall (100%) after preprocessing, outperforming a complex Hybrid CNN–LSTM in sensitivity; and (3) preprocessing improvements yielded comparable or larger performance gains than architectural complexity enhancements. These results challenge the prevailing assumption that more sophisticated deep learning architectures are always necessary for state-of-the-art EEG classification performance, and demonstrate that systematic preprocessing evaluation should precede architectural complexity increases.

5.2. Why Baseline Correction Matters: Mechanistic Insights

5.2.1. Signal Quality Perspective

Baseline shifts (including DC offsets and slow drifts) create artificial signal baselines that distort feature computation. Statistical features such as mean, variance, and power spectral density estimates are directly affected by baseline shifts [29]. Per-channel variability arises because different electrodes have different offsets due to skin–electrode interface variations, referencing choices, and amplifier characteristics, while temporal drift causes offsets to change across windows within a recording. These issues are particularly problematic for portable EEG devices with dry electrodes, which have higher impedance than research-grade wet electrode systems [11, 32]. Standard preprocessing pipelines recognize baseline correction as essential [41], with tools like Brainstorm and EEGLAB providing built-in DC offset removal options [6, 17].

The spectral analysis methods commonly used for EEG feature extraction, including Welch's method for power spectral density and wavelet transforms, are sensitive to baseline shifts [35]. When baseline offsets vary across

Figure 5: Patient-level confusion matrices for both models across conditions. **Top row**: Tabular MLP showing elimination of all false negatives (FN: 2→0) after baseline correction, achieving perfect recall. **Bottom row**: Hybrid CNN–LSTM maintaining perfect precision (no false positives) while reducing false negatives (FN: 3→2). Color intensity reflects cell values.

channels, relative band power calculations (which normalize by total power) can produce misleading values, as the offset contributes differently to different frequency bands. Advanced multichannel EEG analysis tools emphasize the importance of reference-independent topographic analyses precisely because reference and baseline issues can render statistical results ambiguous [7]. Recent work has shown that preprocessing choices, including baseline correction, can substantially shape decoding performance [27], highlighting the need for explicit evaluation of preprocessing effects rather than assuming standard steps are sufficient.

5.2.2. Model Learning Perspective

Without proper baseline correction, models may learn to recognize offset patterns rather than true physiological signals. This creates two problems: (1) the learned features may not generalize across recordings with different offset characteristics, and (2) model capacity is wasted on learning to compensate for preprocessing deficiencies rather than learning discriminative neural patterns. The dramatic improvement in the Tabular MLP, which has limited capacity to learn complex compensatory patterns, suggests that cleaner input data directly translates to better feature quality and classification performance. This aligns with classical machine learning principles: preprocessing

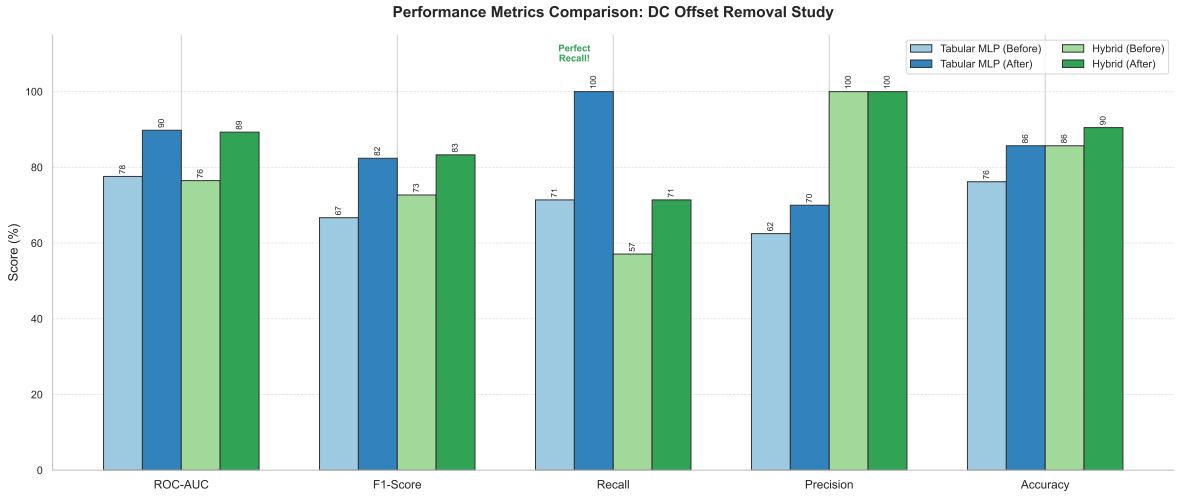

Figure 6: Performance metrics comparison across models and conditions. Grouped bars show ROC-AUC, F1-Score, Recall, Precision, and Accuracy for: Tabular MLP without baseline correction (light blue), Tabular MLP with baseline correction (dark blue), Hybrid baseline (light green), and Hybrid enhanced (dark green). The most dramatic improvement is the Tabular MLP's recall increase from 0.714 to 1.000 (perfect) after baseline correction.

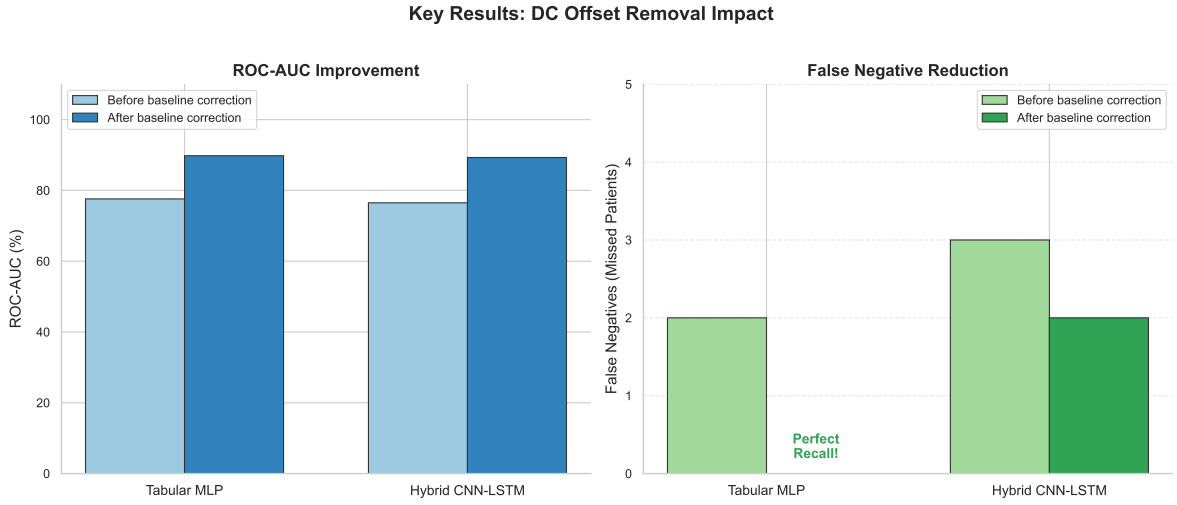

Figure 7: Summary of performance improvements after baseline correction and architectural enhancements. The Tabular MLP (blue) shows larger gains in recall (+40%) and F1-Score (+23.5%) compared to the Hybrid model (green), despite using a simpler architecture. Both models show similar ROC-AUC improvements (∼15–17%), indicating that baseline-correction benefits are broadly architecture-agnostic.

that removes systematic noise allows models to focus on signal rather than artifact.

5.2.3. Feature Computation Stability

Baseline shifts propagate through feature extraction pipelines in multiple ways. First, time-domain statistics (mean, variance, RMS) are directly biased by non-zero baselines. Second, frequency-domain features computed via Welch's method or FFT are affected because the DC component contributes to total power, distorting relative band power calculations. Third, asymmetry features (which compare left vs. right hemisphere power) can be systematically biased if baseline offsets differ across hemispheres. Fourth, normalization steps (e.g., z-scoring or min-max scaling) applied after feature extraction can amplify baseline-induced biases if the baseline varies across windows or channels. Our per-channel, per-window mean-centering addresses these issues by ensuring that each analysis window is centered before feature computation, preventing baseline-induced feature drift that could degrade model calibration and generalization.

5.3. Neurophysiological Interpretation of Selected Features

The feature selection results provide neurophysiological insights consistent with prior work on tDCS mechanisms and depression biomarkers. Four of the top five features originated from the TP10 electrode (right temporal region),

and the fifth captured beta-band asymmetry between left frontal and temporal regions.

5.3.1. Right Temporal Dominance

The prominence of right temporal features aligns with findings on hemispheric imbalance in depression [24]. Depression has been associated with reduced left frontal activity relative to right frontal activity, and tDCS protocols commonly target the left DLPFC with anodal stimulation and right DLPFC with cathodal stimulation to rebalance this asymmetry [36]. The right temporal electrode (TP10) may capture secondary effects of this intervention or reflect baseline differences in patients who respond to treatment.

5.3.2. Beta Band Asymmetry

The selection of left frontal-temporal beta difference as a top feature is consistent with evidence that beta band networks are implicated in depression and treatment response [15]. Beta activity has been linked to cortical inhibition and cognitive control processes that are often disrupted in depression. The asymmetry between frontal and temporal regions may reflect connectivity patterns relevant to mood regulation.

5.3.3. Relation to tDCS Mechanisms

tDCS modulates cortical excitability through subthreshold polarization of neuronal membranes, with effects that persist beyond the stimulation period through neuroplasticity mechanisms [31, 38]. The selected EEG features may capture baseline neural signatures that predict capacity for such neuroplastic changes. Prior work has identified distinct network modulation patterns in tDCS responders versus non-responders [52], and our feature selection results suggest that temporal region activity and frontal-temporal beta connectivity are particularly informative for predicting these differential responses.

5.4. Comparison to Prior Work

Our Tabular MLP with baseline correction achieved 89.8% ROC-AUC with 100% sensitivity and 78.6% specificity. This compares favorably to prior work on the same clinical dataset: Xiao et al. [55] reported 71.9% accuracy (sensitivity 52.9%, specificity 83.1%) using PLV connectivity features with deep learning, and Moncy et al. [33] achieved 85.7% accuracy using end-to-end deep models on PSD features. Our results also exceed the benchmarks from a meta-analysis of 15 EEG-based treatment prediction studies, which reported average sensitivity of 83% and specificity of 86% [13].

Importantly, our approach achieved these results with a substantially simpler model architecture and explicit attention to preprocessing evaluation, whereas prior work emphasized architectural innovations. This suggests that systematic preprocessing evaluation may be a more fruitful avenue for improvement than continued architectural complexity, particularly when working with portable EEG systems where baseline variability may be higher.

5.5. Why Simpler Models Benefit More

The Tabular MLP showed more dramatic recall improvement (0.714 → 1.000) compared to the Hybrid model (0.571 → 0.714). We hypothesize this reflects a **capacity hypothesis**. Simple models have limited capacity to learn complex compensatory patterns for noisy inputs; cleaner data directly improves performance because the model cannot waste capacity on learning to ignore artifacts. In contrast, complex models can partially learn to ignore offsets through their greater representational capacity, but waste capacity doing so; they still benefit from preprocessing but to a lesser degree because some capacity is already allocated to artifact compensation.

This finding has practical implications: rather than investing in increasingly complex architectures, practitioners should first ensure preprocessing quality, which may yield larger improvements with simpler, more interpretable models. The success of the Tabular MLP also aligns with findings from Wu et al. [53], who achieved strong results using SELSER (Sparse EEG Latent Space Regression) with careful feature engineering rather than deep architectures.

The broader implication is that the EEG machine learning literature may be overemphasizing architectural innovation at the expense of preprocessing rigor. When models must compensate for preprocessing deficiencies, their effective capacity for learning discriminative features is reduced, and the "improvements" attributed to architectural changes may partially reflect better artifact tolerance rather than better feature learning.

5.6. Architecture Enhancements: Limited Isolated Impact

The Hybrid model experiments confounded baseline correction with architectural changes (GELU activation, enhanced attention, positional encoding). While overall performance improved substantially, we cannot isolate the contribution of each change. However, comparing the two experiments suggests baseline correction is the primary driver: both models showed ∼0.12–0.13 ROC-AUC improvement, the Tabular MLP improved more in recall despite no architecture changes, and the Hybrid model's improvements are consistent with preprocessing benefits rather than architecture. Future work should include controlled ablations isolating each architectural change to quantify their individual contributions.

5.7. Clinical Implications

The achievement of perfect recall in the Tabular MLP has significant clinical implications for depression treatment selection and the development of reliable biomarkers for predicting tDCS treatment response, which represents a critical step toward precision psychiatry [21]. All patients who would achieve remission are correctly identified, ensuring that potential treatment responders are not denied an effective intervention. The model is suitable for initial screening to identify tDCS candidates,

complementing clinical assessment with objective biomarker data. The presence of 3 false positives (patients predicted to remit who do not) is acceptable in screening contexts where subsequent clinical evaluation would occur. Given tDCS's favorable safety profile [51, 46], the cost of a false positive (patient undergoes ineffective but safe treatment) is lower than the cost of a false negative (patient is denied potentially effective treatment). Early identification of responders could help prioritize tDCS for appropriate patients while directing non-responders sooner to alternative therapies, optimizing healthcare resource utilization.

The Hybrid model's perfect precision (no false positives) suggests a complementary role for confirmation, where high confidence in positive predictions is required. A two-stage approach, screening with the high-recall Tabular MLP followed by confirmation with the high-precision Hybrid model, could balance sensitivity and specificity for clinical decision-making.

5.7.1. Deployment Considerations

The success of the simple Tabular MLP has important implications for clinical deployment. Existing EEG studies often employ high-density electrode systems with 19-128 channels, creating significant barriers to clinical deployment due to complexity, cost, and patient compliance issues [49]. Our results demonstrate that effective prediction is achievable with just 4 electrodes and a simple model, addressing these barriers and facilitating translation from research to practice.

Furthermore, simpler models are easier to interpret, validate, and audit, all critical considerations for clinical applications where regulatory approval and clinician trust are essential. The Tabular MLP's feature-based approach allows examination of which EEG characteristics drive predictions, supporting clinical understanding and model refinement. As EEG-based biomarkers become integrated into cognitive healthcare frameworks that combine IoT devices with deep learning for real-time patient monitoring [4], the importance of reliable preprocessing that works across diverse recording conditions becomes even more critical.

5.8. Limitations

Several limitations must be acknowledged. The Hybrid model experiment confounded baseline correction with architecture changes (GELU activation, enhanced attention, positional encoding), preventing isolation of individual contributions; while the Tabular MLP experiment provides clean evidence for preprocessing effects, the Hybrid model results cannot distinguish preprocessing benefits from architectural improvements. The sample size (n=21 patients, 1,203 windows) limits statistical power and generalizability—while LOPO cross-validation provides conservative, patient-independent estimates that avoid data leakage, the small number of patients (particularly in the remission group, n=7) means that single-patient variations can substantially affect aggregate metrics, and validation on larger, multi-site datasets is essential before clinical deployment [14]. Results are specific to tDCS remission prediction in MDD using home-based EEG acquisition, and generalization to other EEG classification tasks (e.g., rTMS response, antidepressant response), treatment modalities, recording protocols (e.g., research-grade EEG systems), or clinical populations requires additional validation. The feature selection was performed using SelectKBest with ANOVA F-statistic, which assumes features are independent and may not capture nonlinear relationships, and the stability of selected features across different patient subsets was not formally assessed.

5.9. Future Work

Several directions warrant investigation. Ablation studies should isolate architecture contributions in the Hybrid model through controlled factorial experiments testing each component independently. Multi-site validation across different clinical sites, recording equipment, and patient populations is essential to assess generalizability [14]. Future work should investigate learned preprocessing strategies that adapt per channel or patient, potentially improving on fixed mean-subtraction approaches. Validation across different classification problems (e.g., antidepressant response, seizure detection, sleep staging) would establish the generality of the preprocessing-first principle. Multimodal integration combining EEG biomarkers with structural MRI or clinical variables would test whether preprocessing benefits extend to multimodal prediction frameworks [19]. Finally, prospective clinical trials using the developed models for treatment selection would evaluate real-world clinical utility.

6. Conclusion

This study demonstrates that per-channel, per-window baseline correction (mean-centering) provides substantial performance gains (15–17% ROC-AUC improvement) comparable to or exceeding architectural complexity enhancements for EEG-based depression remission prediction. Using EEG data from a randomized controlled trial of home-based tDCS for MDD [51], we showed that a simple Tabular MLP achieved perfect recall (1.0) after baseline correction, eliminating all false negatives and matching the ROC-AUC of a substantially more complex Hybrid CNN–LSTM architecture with approximately 3.5 times as many parameters.

6.1. Key Findings

Baseline correction is a standard preprocessing step that should be explicitly evaluated rather than assumed sufficient. The 15–17% ROC-AUC improvement from per-window mean-centering demonstrates that systematic preprocessing evaluation can yield gains comparable to architectural innovations. Preprocessing quality enables simpler models to excel: the Tabular MLP outperformed the Hybrid CNN–LSTM in recall despite having fewer parameters and less representational capacity. Simple

models can achieve clinical-grade performance: the Tabular MLP's perfect recall (100% sensitivity) exceeds meta-analytic benchmarks of 83% sensitivity [13], demonstrating that sophisticated architectures are not prerequisites for excellent performance. Clinical deployment is facilitated by simpler models, which are easier to interpret, validate, audit, and deploy in clinical settings where regulatory approval and clinician trust are essential.

6.2. Recommendations for Practitioners

Based on our findings, we offer the following recommendations for EEG machine learning practitioners. First, prioritize preprocessing evaluation: before investing in complex architectures, systematically evaluate preprocessing pipelines including explicit baseline correction (e.g., per-window mean-centering), artifact rejection, and channel-specific quality assessment. Second, start simple: begin with interpretable models (e.g., tabular classifiers, logistic regression) to establish baselines, and only add architectural complexity if preprocessing-optimized simple models are inadequate. Third, document preprocessing fully by reporting all preprocessing steps in sufficient detail for replication, including baseline correction methods, filter specifications, and artifact rejection criteria. Fourth, conduct ablation studies: when combining preprocessing and architecture changes, isolate contributions through controlled factorial experiments to identify the true sources of improvement. Finally, validate on portable EEG systems, as preprocessing is especially critical for portable and home-based EEG systems, which may have higher baseline variability than research-grade systems.

6.3. Broader Implications

These findings challenge the prevailing assumption in the EEG machine learning literature that increasingly complex architectures are necessary for state-of-the-art classification. The field's emphasis on novel architectures may have led to underinvestment in preprocessing optimization, which our results suggest can yield larger and more reliable improvements. This has implications for reproducibility: when preprocessing varies across studies, "improvements" attributed to architectural changes may partially reflect preprocessing differences.

For precision psychiatry, our results support the feasibility of EEG-based biomarkers for personalizing depression treatment [21]. The achievement of perfect recall with a simple model and portable EEG suggests that tDCS response prediction could be integrated into clinical practice without requiring specialized equipment or computational infrastructure.

In conclusion, baseline correction is a fundamental preprocessing step that should be explicitly evaluated rather than assumed sufficient. Our demonstration that systematic preprocessing evaluation can transform classification performance, enabling a simple model to match complex architectures and achieve clinically relevant sensitivity, underscores the importance of data quality over model complexity. We encourage the field to rebalance its attention toward systematic preprocessing evaluation as a first-order priority for improving EEG-based biomarker development, particularly when working with portable EEG systems where baseline variability may be substantial.

CRediT authorship contribution statement

Abin Daniel Zorto: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.. **Jijomon C. Moncy:** Writing - review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization.. **Mhd Saeed Sharif:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Cynthia H.Y. Fu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

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